ITAI 2372 · Conceptual Design Track

NASA Space Mission AI Project

AI-Powered Autonomous Fault Detection & Recovery System for Deep-Space Spacecraft

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What is the Problem?

Far Away

Spacecraft in deep space can be
millions of miles from Earth.
Signals take hours to travel.

Things Break

Radiation and extreme cold can damage spacecraft parts at any time without warning.

No Human Help

Astronauts or ground teams cannot fix problems instantly when there is a huge delay.

Our AI Solution

AFDRS – Autonomous Fault Detection & Recovery System

WATCH

Reads thousands of sensor readings every second from all spacecraft systems

DETECT

Notices when something looks wrong or unusual — like a smoke alarm for spacecraft

DIAGNOSE

Figures out WHAT is broken and HOW serious the problem is

FIX

Automatically takes safe action to fix the problem or prevent it getting worse

What Data Does the AI Watch?

The AI monitors 6 key spacecraft systems — every 100 milliseconds!

Power

Battery level
Solar panel output
Power per device

Temperature

Heat on all parts
Cooling system
Thermal readings

Propulsion

Engine thrust
Fuel pressure
Valve positions

Comms

Signal strength
Transmission errors
Antenna direction

Life Support

Oxygen levels
CO_2 levels
Cabin pressure

Navigation

Gyroscope data
Reaction wheels
Thruster firings

How Does the AI Work?

3 Smart Techniques — in Simple Words

1

Isolation Forest

"Spots the odd one out"

Like a teacher who notices one student acting differently from the whole class — the AI finds sensor readings that look unusual.

2

LSTM Neural Network

"Remembers normal patterns"

Like knowing your friend usually texts back in 5 minutes — if they suddenly take 2 hours, you notice. The AI learns normal behavior over time.

3

Random Forest

"Takes a vote to decide"

Like asking 100 doctors for their opinion and going with the majority. The AI uses many decision trees to classify what type of fault occurred.

How It Works — Step by Step

1

MONITOR

AI reads all sensors every 0.1 seconds — 24 hours a day

2

DETECT

Something looks wrong! AI flags the unusual reading

3

DIAGNOSE

AI identifies WHAT failed and HOW serious it is

4

DECIDE

If 85%+ confident → fix automatically. Otherwise → alert crew

5

LOG

All events are saved so NASA engineers can review later

Hardware & Software Needed

HARDWARE

Radiation-hardened CPU

Survives space radiation

32 GB ECC Memory

Stores sensor history

1 TB Solid State Drive

Saves all mission logs

Real-Time Operating System

Responds in milliseconds

SOFTWARE

Python

Design and train AI models

TensorFlow Lite

Runs AI on small spacecraft computer

C++

Fast execution in real-time

NASA GMAT Tool

Simulate and test the mission

How We Test the AI

Phase 1

Unit Testing

- Test each module alone
- Inject fake faults into data
- Check each one works correctly

Phase 2

Full System Test

- Connect all 4 modules
- Run simulated missions
- Test how they work together

Phase 3

Stress Testing

- Break 2 things at once
- Turn off some sensors
- Test worst-case scenarios

GOALS: Detect faults in < 0.5 seconds | 95%+ accuracy | Less than 5% false alarms | Fix 90%+ of problems

Expected Results & Impact

What this AI system will achieve for NASA missions

< 0.5s	95%+	90%+	24 / 7
Fault detection speed	Accuracy in finding faults	Problems fixed automatically	Always on, no fatigue

Real-World Impact

🚀 Safer missions — problems caught before they become dangerous

💰 Saves money — fewer failed missions and emergency repairs

🌌 Enables further exploration — Mars, asteroids, and beyond

🤖 Reusable — same system can work on any future NASA spacecraft

Thank You!

Project Summary

- Problem: Spacecraft faults with no quick human help
- Solution: AI that watches, detects, diagnoses, and fixes
- Track: Conceptual Design (detailed plan, no coding)
- AI Used: Isolation Forest + LSTM + Random Forest
- Testing: Unit → Integration → Stress Testing
- Goal: Safer, smarter, fully autonomous space missions

References

- NASA Technical Reports Server (ntrs.nasa.gov)
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- Isolation Forest — Liu et al., IEEE 2008
- LSTM Networks — Hochreiter & Schmidhuber, 1997
- Random Forests — Breiman, Machine Learning 2001
- Reinforcement Learning — Sutton & Barto, 2018